

Probing Vision–Language–Action Encoders for Affordance

Class-asymmetric degradation, mechanism, and recovery

Nitik Jain

EN.601.495 / 695 Introduction to Robot Learning Spring 2026

Motivation

Manipulation requires **part-level** affordance perception.

Handle vs. blade. Rim vs. bottom. Lever vs. shaft.

The literature establishes that adding affordance *helps* manipulation:

Aff-Grasp (ICCV 2025)	+30 pp grasp success
TARAD (RA-L 2024)	+52 pp task success
GIFT (RSS 2021)	+22 pp novel tools
GRAFF (ICRA 2021)	3× faster RL
FSD (ICLR 2026)	+30 pp real-world

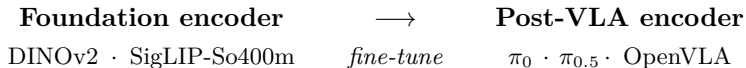
We do not re-prove this. We cite it. Our work is upstream.



UMD Part Affordance: per-pixel labels of grasp / cut / scoop / contain / support.

The problem

Modern policies are vision-language-action models (π_0 , $\pi_{0.5}$, OpenVLA). Each starts from a foundation vision encoder and **fine-tunes it end-to-end** on robot trajectories.



Does the post-VLA encoder retain the per-class affordance signal that its foundation encoder had?

No published study isolates this — not the per-class structure of the loss, nor whether the loss is recoverable.

Prior work, in one slide

Direction	Status
Affordance-conditioned manipulation Aff-Grasp, TARAD, GIFT, GRAFF, FSD	established +20–52 pp success
Affordance probing on <i>standalone</i> encoders Zhang et al. CVPR 2026, dense decoder, UMD	established 0.670 mIoU
Per-class affordance loss <i>inside</i> a deployed VLA encoder	gap
Per-class loss $\rightarrow$ mechanism $\rightarrow$ recovery	gap

Two well-developed prior literatures, one untouched gap. We close the latter.

Key insight

If VLA fine-tuning **rotates** per-class affordance directions out of a single readable hyperplane, rather than **deleting** them, a small adapter on *frozen* features should recover the signal.

Falsifiable. Linear probing alone cannot tell rotation from deletion. An MLP adapter on frozen features can.

Method

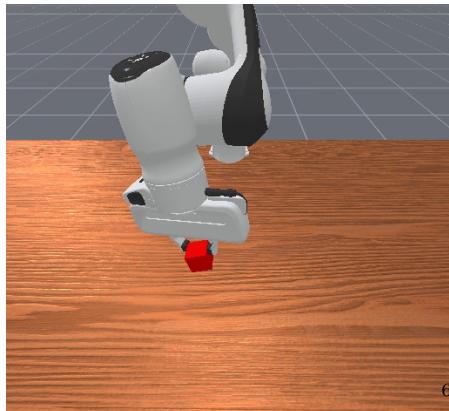
Pipeline. VLA checkpoint $\rightarrow$ locate vision-tower keys $\rightarrow$ load into HF / timm skeleton $\rightarrow$ frozen forward $\rightarrow$ *linear probe* (60 s sklearn LR) or **MLP adapter** (1152 $\rightarrow$ 256 $\rightarrow$ 6, 297K params).

Encoders evaluated (7 encoders, 2 VLA families)

- ▶ Vision-only: DINOv2-base, DINOv2-large
- ▶ Foundation: SigLIP-So400m
- ▶ PaliGemma family: π_0 , $\pi_{0.5}$ SigLIP
- ▶ Prismatic-Llama family: OpenVLA SigLIP
- ▶ Intervention: π_0 +adapter, $\pi_{0.5}$ +adapter, OpenVLA+adapter

Probe protocol. Frozen patch features $\rightarrow$ multinomial LR with $L_2 = 1$.

Patch labels = majority of pixel labels under each ViT 14 $\times$ 14 patch.



Experimental design *(a complexity spectrum)*

Test	Setup	Difficulty
H2	5-class IoU on full UMD	hardest
H10a	binary cut vs. rest-foreground, full UMD	
H10b	multi-tool composite scenes (knife + distractor)	
H9	binary handle vs. blade, knife / scissors only	easiest

Different prior probes use different protocols. We span the spectrum to localise where the loss *actually* lives.

7 encoders $\times$ 4 tasks. UMD train/val/test = 345 / 73 / 75 single-tool, 22 composites. RTX 5060 Laptop, 8 GB VRAM, no cloud spend.

Results (1) probe baseline (2) class-asymmetric loss

0.776

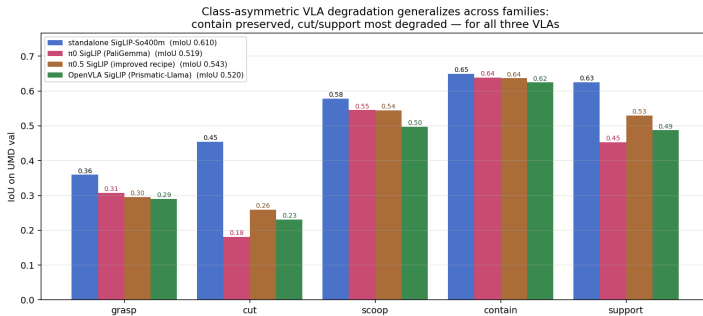
mIoU on UMD val

DINOv2-large @ 560²,
60-second linear probe.

Zhang et al. CVPR 2026:
0.670

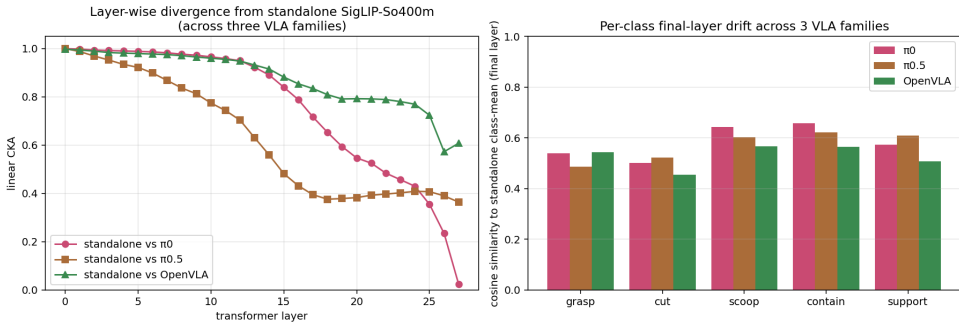
+10.6 pp

no decoder, no fine-tuning.



cut -27 pp | **contain -1 pp** | *pattern persists across
PaliGemma and Prismatic-Llama VLA families.*

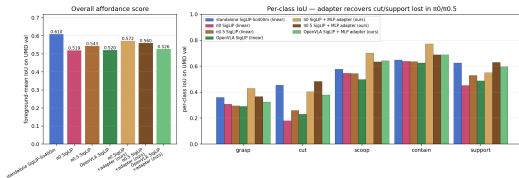
Mechanism *where in the encoder does the loss live?*



π_0 keeps the encoder almost intact through layer 12, then **rotates the final layer to CKA = 0.02**.
Per-class final-layer drift predicts per-class IoU drop: Spearman $\rho = 0.90$, $p = 0.037$ (π_0); Pearson $r = 0.95$, $p = 0.012$ (OpenVLA).

Intervention 297K-parameter adapter on frozen features

Frozen-backbone adapter recovery: VLA-degraded affordance is recoverable, not deleted

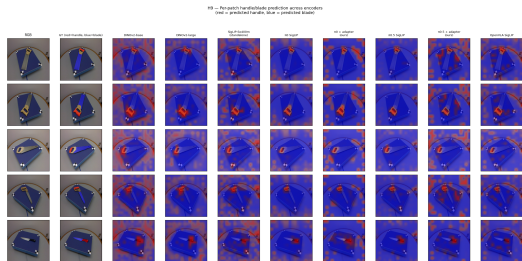


backbone / head	val mIoU	cut IoU
standalone / linear	0.610	0.455
π_0 / linear	0.519	0.181
$\pi_{0.5}$ / linear	0.543	0.259
OpenVLA / linear	0.520	0.231
π_0 / adapter	0.642	0.405
$\pi_{0.5}$ / adapter	0.632	0.483
OpenVLA / adapter	0.604	0.379

82% of the cut-class gap closed on π_0 .

π_0 +adapter *exceeds* the standalone-linear ceiling.

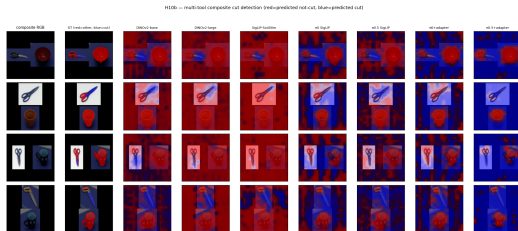
Qualitative results



H9: per-patch handle / blade prediction across encoders.

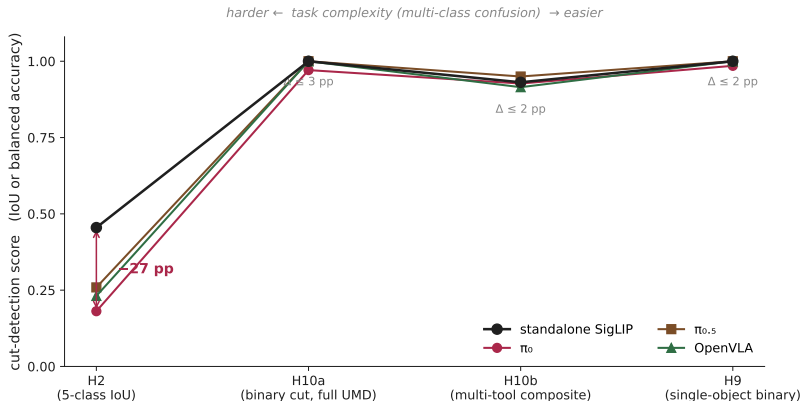
Red = predicted handle; blue = predicted blade.

Every encoder — including the VLAs we said *lost* cut-class — correctly localises blade vs. handle on real UMD knives, and that performance survives placing a distractor in the same scene.



H10b: composite scenes (knife + distractor). Across all encoders the predicted cut region lands on the blade.

Analysis *when does the cut-class loss actually appear?*



π_0 -standalone: H2 = -27 pp | H10a = -3 pp | H10b = -0.4 pp | H9 = -1.5 pp.

Expected: the -27 pp deficit to bite on every cut-related test. *Found:* it collapses on every formulation a real manipulation pipeline uses.

Failure cases & limitations

- ▶ Adapter recovers per-pixel UMD probe but **does not transfer** cleanly to multi-tool composites.
 π_0 +adapter on H10b drops to 0.85 vs. raw π_0 at 0.93. The adapter is dataset-specific.
- ▶ No closed-loop manipulation evaluation.
 $\text{LIBERO} \times \pi_0$ requires ≥ 24 GB VRAM for π_0 inference; we had 8. Cloud GPU is the natural follow-up.
- ▶ Complexity spectrum has only **four** points.
Adding AGD20K, IIT-AFF, RGB-D Part Affordance would tighten the conclusion.
- ▶ Mechanism correlation, $n = 5$ classes per VLA.
Significant for 2 of 3 VLAs. A wider taxonomy would let us claim more.

Conclusion

1. VLA fine-tuning **rotates** per-class affordance directions class-asymmetrically. **Cut** most, **contain** least.
2. The rotation is **mechanistically localised**: per-class drift predicts per-class IoU drop with $\rho = 0.90$ (π_0) and $r = 0.95$ (OpenVLA).
3. A **297K-parameter adapter** on frozen features recovers 82% of the cut-class gap.
4. The standard multi-class probe **overstates** the deficit. Binary part-discrimination — the formulation manipulation actually uses — is preserved.

Future work: $\text{LIBERO} \times \pi_0$ per-task success-rate prediction from per-class probe scores.

Code, weights, and figures: available on request.